

System Model

Microgrid Architecture

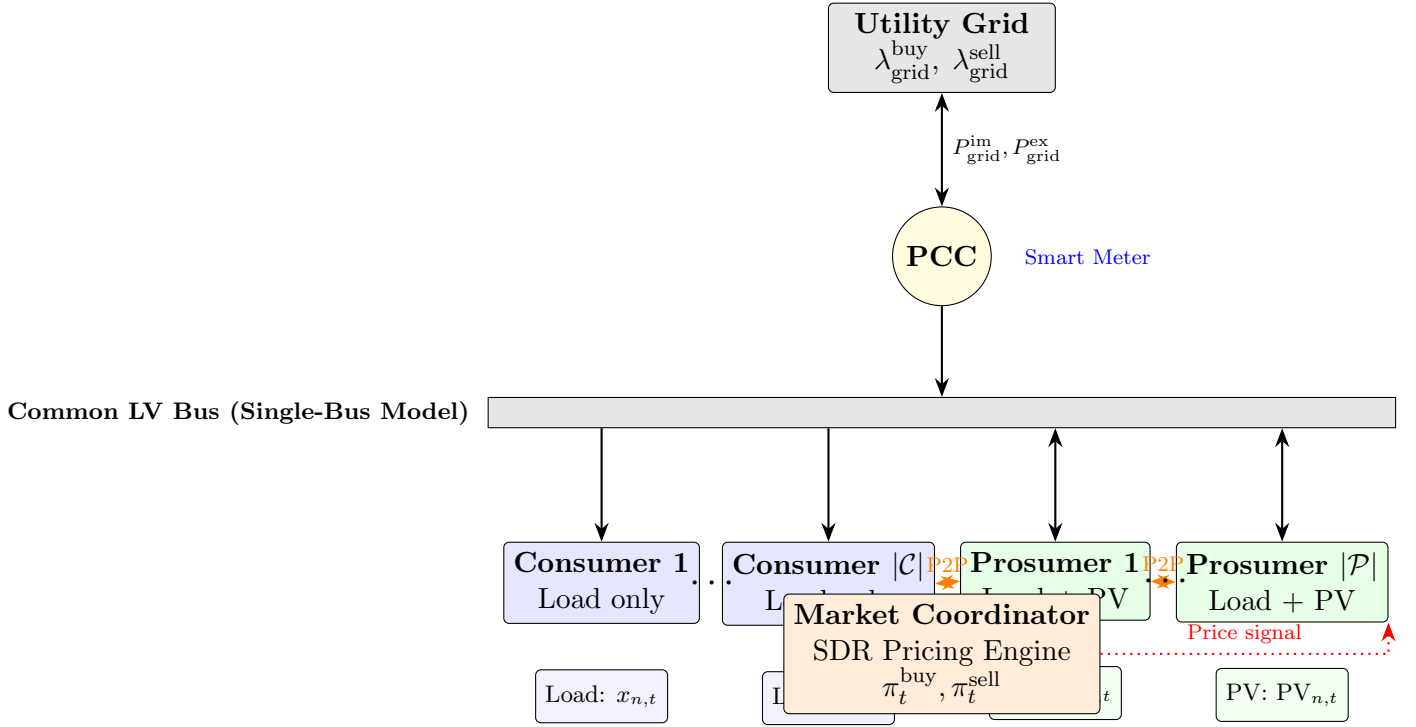
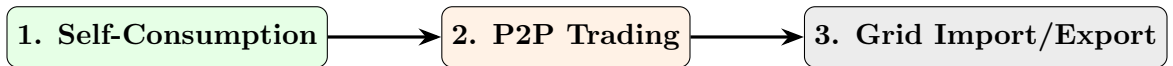


Figure 1: *

Fig. 1: Single-bus microgrid model. All N households connect to a common LV bus via smart meters. The Point of Common Coupling (PCC) interfaces with the utility grid. P2P energy sharing occurs internally; residual imbalance is settled with the grid at regulated tariffs. The market coordinator computes SDR-based clearing prices each slot.

Energy Flow Priority



Each prosumer first consumes own PV generation. Surplus is offered to the P2P market. Only the residual (unmatched P2P demand/supply) interacts with the utility grid.

Simulation Assumptions

Category	Assumptions
1. Network Model	• Single-bus (copper-plate) model — all households on a common LV bus • Lossless internal power transfer (line losses neglected) • No voltage, thermal, or power flow constraints modeled • Valid for small LV communities where feeder losses <2% [following Liu et al., 2017]
2. Metering & Communication	• Smart meters at each household with 15-minute resolution ($\Delta t = 0.25$ h) • Instantaneous measurement and reporting — no communication delays • Perfect metering accuracy (no measurement error modeled)
3. P2P Trading Rules	• Priority: Self-consumption $\rightarrow$ P2P trading $\rightarrow$ Grid interaction • P2P energy allocated proportionally to each participant's surplus/deficit • No transaction fees, network-use charges, or minimum trade volumes • Internal prices bounded: $\pi_{\text{grid}}^{\text{sell}} \leq \pi_t^{\text{int}} \leq \pi_{\text{grid}}^{\text{buy}}$ (no-arbitrage)
4. Market Clearing	• Instantaneous clearing at each time slot (no settlement delays or queuing) • Price-taking behavior — no strategic bidding or market power • Perfect information: all participants observe the same clearing price • Community power balance: $\sum_n P_{n,t}^{\text{P2P,buy}} = \sum_n P_{n,t}^{\text{P2P,sell}} \forall t$
5. Load Profiles	• Residential dual-peak load shape (morning $\sim$8:00, evening $\sim$20:00) • Consumer daily load: 4–7 kWh; Prosumer daily load: 5–9 kWh • Day-to-day variability: Gaussian noise ($\sigma = 5\%$ daily scaling, 5% per-slot)
6. PV Generation	• Clear-sky sinusoidal profile (06:00–18:00), peaking at solar noon • PV yield: 4.5–5.5 kWh/kWp/day (typical Indian conditions) • Daily cloud variability: Gaussian ($\sigma = 15\%$) • Consumers have zero PV capacity
7. DSM Constraints	• Load bounds: $D_n^{\min} \leq x_{n,t} \leq D_n^{\max}$ (set at $\pm 50\%$ of $D_{n,t}^{\text{ref}}$) • Daily energy conservation: $\sum_{t \in d} x_{n,t} \Delta t = \sum_{t \in d} D_{n,t}^{\text{ref}} \Delta t$ • Iterative price-response with damping ($\gamma = 0.3$) for convergence
8. Grid Tariffs	• Flat (time-invariant) tariffs: $\lambda_{\text{grid}}^{\text{buy}} = \text{Rs } 6.0/\text{kWh}$, $\lambda_{\text{grid}}^{\text{sell}} = \text{Rs } 3.4/\text{kWh}$ • No time-of-use or dynamic grid pricing • Same buy price for consumers and prosumers
9. Fairness Evaluation	• $\mathcal{F} = 1 - \frac{ s_C - s_P }{ s_C + s_P + \varepsilon}$ where s = group savings % vs conventional • Post-hoc metric (not a constraint in the optimization) • $\mathcal{F} = 1$: equal savings for both groups; $\mathcal{F} = 0$: one group gains nothing